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Trap device and semiconductor manufacturing device

Abstract

A trap device including an exhaust gas introduction pipe configured to allow an exhaust gas to flow and be led out from an outlet; a fin member provided in a position where the fin member faces the outlet and is hit by the exhaust gas led out from the outlet; an exhaust path forming member covering at least a portion of the exhaust gas introduction pipe and including an exhaust path configured to exhaust the exhaust gas such that the exhaust gas is turned back, via the fin member, in an opposite direction to a direction of flow of the exhaust gas in the exhaust gas introduction pipe; and a cooling jacket configured to cool the fin member is provided. The fin member includes a fin extending in the opposite direction to the direction of the flow of the exhaust gas in the exhaust gas introduction pipe.

Inventors:	Furuya; Yuichi (Yamanashi, JP), Sakabe; Hiroki (Yamanashi, JP)
Applicant:	Tokyo Electron Limited (Tokyo, JP)
Family ID:	1000008780158
Assignee:	Tokyo Electron Limited (Tokyo, JP)
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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1514441	12/1923	Culp	55/417	F01N 1/084
3483980	12/1968	Cochran	210/186	G21C 19/313
5405445	12/1994	Kumada	118/724	C23C 16/4412
5820641	12/1997	Gu	55/DIG.15	B01D 53/002
6045618	12/1999	Raoux	55/284	B01D 45/06
6156107	12/1999	Hayashi	438/905	B01D 45/08
6187072	12/2000	Cheung	118/715	C23C 16/4405
6193802	12/2000	Pang	118/723R	H01J 37/32844
6197119	12/2000	Dozoretz	55/525	H01J 37/32844
6238514	12/2000	Gu	118/715	H01J 37/32844
6303006	12/2000	Chang	202/205	B01D 3/10
6488745	12/2001	Gu	55/DIG.15	B01D 8/00
7044997	12/2005	Mardian	96/417	B01D 5/0006
7727296	12/2009	Tojo	438/905	B01D 45/06
10036090	12/2017	Komori	N/A	C23C 16/18
10369506	12/2018	Hakki	N/A	B01D 53/504
2001/0003892	12/2000	Rikyuu	55/444	C23C 16/4412
2003/0198578	12/2002	Lee	422/138	C23C 16/452
2006/0162862	12/2005	Park	N/A	N/A
2011/0045182	12/2010	Fukumori	118/724	C23C 16/4412
2021/0291222	12/2020	Nishi	N/A	H01L 21/67017
2023/0360895	12/2022	Furuya	N/A	C23C 16/4412

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
H11-300153	12/1998	JP	N/A
2004-305950	12/2003	JP	N/A

Primary Examiner: Zervigon; Rudy

Attorney, Agent or Firm: IPUSA, PLLC

Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to trap devices and semiconductor manufacturing devices.

BACKGROUND ART

(2) For example, Patent Document 1 proposes an exhaust trap in which a trap container with an inlet and outlet for an exhaust gas houses a filtration-media-filled filter with a cross-sectional area larger than the inlet, and a space is provided between the inlet and the filtration-media-filled filter. The space between the inlet and the filtration-media-filled filter allows the exhaust gas that is introduced from the inlet to reach the filtration-media-filled filter after having spread in the space. As a result, it is possible to reduce clogging of the filtration-media-filled filter, and thus prolong the life of the exhaust trap.

(3) For example, Patent Document 2 proposes a substrate processing apparatus that includes an exhaust system configured to exhaust gas without byproducts by cooling the inside of a trap provided in the exhaust system to reduce the temperature of the exhaust gas.

PRIOR ART DOCUMENT

Patent Document

(4) [Patent Document 1] Japanese Laid-Open Patent Application Publication No. 2004-305950

(5) [Patent Document 2] Japanese Laid-Open Patent Application Publication No. 2009-16635

SUMMARY OF THE INVENTION

Problem to be Solved by the Invention

(6) The present disclosure provides a trap device and a semiconductor manufacturing device capable of increasing the entrapment efficiency of byproducts contained in an exhaust gas.

Means for Solving the Problem

(7) According to one aspect of the present disclosure a trap device includes an exhaust gas introduction pipe configured to allow an exhaust gas to flow and be led out from an outlet; a fin member provided in a position where the fin member faces the outlet and is hit by the exhaust gas led out from the outlet; an exhaust path forming member covering at least a portion of the exhaust gas introduction pipe and including an exhaust path configured to exhaust the exhaust gas such that the exhaust gas is turned back, via the fin member, in an opposite direction to a direction of flow of the exhaust gas in the exhaust gas introduction pipe; and a cooling jacket configured to cool the fin member, wherein the fin member includes a fin extending in the opposite direction to the direction of the flow of the exhaust gas in the exhaust gas introduction pipe.

Effects of the Invention

(8) According to one aspect, it is possible to increase the entrapment efficiency of byproducts contained in an exhaust gas.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic cross-sectional view illustrating an example of a semiconductor

manufacturing device according to the first and second embodiments;

(2) FIG. 2 is a vertical cross-sectional view of a trap device according to the first embodiment;

(3) FIG. 3A is a view illustrating a fin member in the trap device according to the first embodiment;

(4) FIG. 3B is a view illustrating the fin member in the trap device according to the first embodiment;

(5) FIG. 4 is a view of an exhaust path in the trap device taken along a B-B plane of FIG. 2;

(6) FIG. 5 is a vertical cross-sectional view of the trap device according to the second embodiment;

(7) FIG. 6A is a view illustrating the fin member in the trap device according to the second embodiment; and

(8) FIG. 6B is a view illustrating the fin member in the trap device according to the second embodiment.

DESCRIPTION OF THE EMBODIMENTS

(9) Embodiments of the present disclosure will be described hereinafter with reference to the accompanying drawings. Note that the same reference numerals are used to denote the same components throughout the drawings, and a repetitive description thereof will be omitted.

(10) [Semiconductor Manufacturing Device]

(11) A semiconductor manufacturing device **10** according to the first and second embodiments will be described first with reference to FIG. 1. FIG. 1 is a schematic cross-sectional view illustrating an example of the semiconductor manufacturing device **10** according to the first and second embodiments. FIG. 1 illustrates a capacitively coupled plasma chemical vapor deposition (CVD) apparatus.

(12) The semiconductor manufacturing device **10** includes a chamber **1**, and a processing space **1s** where deposition processes and etching processes are performed is provided in the chamber **1**. In the semiconductor manufacturing device **10**, plasma is formed between an upper electrode **3** and a stage **ST** in the chamber **1**, and a substrate **W** is processed by the plasma acting on the substrate **W**. The stage **ST** includes a lower electrode **4** and an electrostatic chuck **5**. The stage **ST** may include a heater. The substrate **W** is held above the lower electrode **4**. An RF power supply **6** and an RF power supply **7** are coupled to both the upper electrode **3** and the lower electrode **4**, and different RF frequencies can be used. Alternatively, the RF power supply **6** and the RF power supply **7** may be coupled to the same electrode. Further, a direct current (DC) power supply may be coupled to the upper electrode **3**. The chamber **1** is connected to a gas supply **8**, and processing gas is supplied to the processing space **1s**. An exhaust device **9** is also connected to the chamber **1** and exhausts the air in the chamber **1**. A trap device **20** is provided in a gas exhaust line **L** that is disposed between the chamber **1** and the exhaust device **9**, and the trap device **20** traps byproducts contained in the exhaust gas. The trap device **20** may be provided, for example, between a turbomolecular pump and a dry pump that serve as examples of the exhaust device **9**.

(13) The semiconductor manufacturing device **10** includes a controller **40** that includes a processor and a memory. The controller **40** controls each component of the semiconductor manufacturing device **10** to perform plasma processing such as deposition processes on the substrate **W**.

First Embodiment

(14) [Trap Device]

(15) The configuration of the trap device **20** according to the first embodiment will be described next with reference to FIGS. 2 to 4. FIG. 2 is a vertical cross-sectional view of the trap device **20** according to the first embodiment. FIGS. 3A and 3B are views each illustrating a fin member **23** in the trap device **20** according to the first embodiment. FIG. 3A is a plan view of the fin member **23** according to the first embodiment and is taken along a plane A-A of FIG. 2. FIG. 3B is a perspective view of the fin member **23** according to the first embodiment. FIG. 4 is a cross-sectional view taken along a plane B-B of FIG. 2 and is a view illustrating the periphery of a second exhaust path **33** in the trap device **20**.

(16) The trap device **20** is a device configured to resolidify and trap byproducts contained in an

exhaust gas that is exhausted when processing gas is used to perform deposition or etching on the substrate W in the processing space 1s of the chamber 1. The byproducts contained in the exhaust gas include, for example, precursor materials that were not used in deposition. For example, in a case where a ruthenium film is deposited on the substrate W by using a Ru.sub.3(CO).sub.12 process gas (precursor gas), byproducts such as ruthenium and the like are contained in the exhaust gas. In such a case, the trap device 20 traps the ruthenium and the like contained in the exhaust gas. (17) Releasing the byproducts contained in the exhaust gas directly into the atmosphere causes environmental pollution. Further, depending on the type of the exhaust gas, deposits may be generated inside the exhaust system. This may, for example, reduce the conductance of the gas exhaust line L or cause the exhaust device 9 to break down.

(18) Hence, to exhaust the process gas that was not used in deposition, the trap device 20 is provided in an exhaust system such as the gas exhaust line L. The exhaust gas flowing in the trap device 20 is cooled to resolidify the byproducts, and the resolidified byproducts are trapped. Therefore, the trap device 20 according to the embodiment focuses on the flow of the exhaust gas in the trap device 20 to improve the efficiency of trapping byproducts, and thus a configuration that can create a flow of exhaust gas to trap byproducts efficiently is proposed.

(19) More specifically, as illustrated in FIG. 2, the trap device 20 includes an exhaust gas introduction pipe 21, an exhaust path forming member 22, the fin member 23, a cooling jacket 24, and a housing 25. The exhaust gas introduction pipe 21 has an L-shaped vertical cross-section, and includes a pipe 21a, a pipe 21b, an inlet 30, and an outlet 32. The pipe 21a includes the inlet 30 for the exhaust gas, and the pipe 21b includes the outlet 32 for the exhaust gas.

(20) The pipe 21a extends in a vertical direction, and the inlet 30 opens upward at one end of the pipe 21a. The other end of the pipe 21a is connected to one end of the pipe 21b that is provided substantially perpendicular to the pipe 21a. The pipe 21b extends in a horizontal direction, and the outlet 32 opens sideways in the other end of the pipe 21b. The exhaust gas is introduced to the exhaust gas introduction pipe 21 from the inlet 30 via the gas exhaust line L.

(21) The pipe 21a and the pipe 21b are cylindrical, and the length of the pipe 21b is longer than the length of the pipe 21a. The central axis of the pipe 21b substantially matches a central axis Ax of the trap device 20. Assume hereinafter that “substantially matches” includes deviations due to the precision of machining.

(22) The exhaust gas introduction pipe 21 is preferably made of metal such as aluminum, an aluminum alloy, or stainless steel. The exhaust gas introduction pipe 21 includes a heater and is able to heat the exhaust gas that passes through the exhaust gas introduction pipe 21. The exhaust gas passes a first exhaust path 31 in the exhaust gas introduction pipe 21 through the inlet 30, and is led out into the trap device 20 from the outlet 32. The exhaust gas is heated as it passes the first exhaust path 31.

(23) The other end (the outlet 32) of the exhaust gas introduction pipe 21 is inserted into the housing 25, and the one end (the inlet 30) of the exhaust gas introduction pipe 21 is exposed from the housing 25. A ring-shaped protrusion 21c is formed on the exposed portion of the exhaust gas introduction pipe 21. An O-ring 38 is provided between the protrusion 21c and an engagement portion 25a that is provided on the one end of the housing 25 and extends along the outer periphery of the exhaust gas introduction pipe 21. The O-ring 38 keeps the housing 25 airtight and prevents the exhaust gas and the resolidified byproducts from leaking outside of the trap device 20 from between the exhaust gas introduction pipe 21 and the housing 25. An exhaust vent 37 for the exhaust gas is formed on the other end of the housing 25.

(24) The housing 25 is preferably made of metal such as aluminum, an aluminum alloy, or stainless steel. The cooling jacket 24 is provided externally to the housing 25 and is provided along the housing 25 around the exhaust vent 37. The cooling jacket 24 includes a flow path 24a, and the housing 25, the exhaust path forming member 22, and the fin member 23 are cooled by a refrigerant such as a coolant that flows in the flow path 24a.

(25) The housing **25** is cylindrical and includes the exhaust path forming member **22**. The exhaust path forming member **22** is cylindrical and is of a diameter smaller than that of the housing **25**. The exhaust path forming member **22** is preferably made of aluminum, an aluminum alloy, or stainless steel. The exhaust path forming member **22** is fixed to the housing **25**.

(26) The diameter of the exhaust path forming member **22** is larger than the diameter of the pipe **21b**, and the exhaust path forming member **22** is disposed so as to cover at least a portion of the pipe **21b**. The central axes of the pipe **21b**, the exhaust path forming member **22**, and the housing **25** substantially coincide with the central axis Ax. That is, the pipe **21b**, the exhaust path forming member **22**, and the housing **25** are arranged concentrically about the central axis Ax.

(27) The exhaust path forming member **22** is provided between the pipe **21b** and the housing **25**, and covers the pipe **21b** from the side where the outlet **32** is inserted in the housing **25**. As a result, a second exhaust path **33**, which is defined by the outer surface of the pipe **21b** and the inner surface of the exhaust path forming member **22**, is formed between the pipe **21b** and the exhaust path forming member **22**. A third exhaust path **34**, which is defined by the outer surface of the exhaust path forming member **22** and the inner surface of the housing **25**, is formed between the exhaust path forming member **22** and the housing **25**. The first exhaust path **31**, the second exhaust path **33**, and the third exhaust path **34** are formed from the inner side in this order in the horizontal direction.

(28) The exhaust path forming member **22** is shaped like a cylinder with a base. The fin member **23** is attached to the base of the exhaust path forming member **22**. The fin member **23** attached to the base of the exhaust path forming member **22** is disposed to face the outlet **32** of the pipe **21b** such that the exhaust gas led out from the outlet **32** hits the fin member **23**. For example, the fin member **23** is disposed in a position where the exhaust gas led out from the outlet **32** hits the fin member **23** at a substantially right angle.

(29) The fin member **23** includes fins **26** that extend in an opposite direction to the direction of the flow of the exhaust gas in the exhaust gas introduction pipe **21**. FIG. 3A is a plan view of the fins **26** taken along a plane A-A of FIG. 2. Further, FIG. 3B is a perspective view of the fins **26**.

(30) The fins **26** include three fins **26a**, **26b**, and **26c**. These fins **26a**, **26b**, and **26c** may be referred to collectively as the fins **26** hereinafter. The fins **26a**, **26b**, and **26c** are ring-shaped members with different diameters from each other, are arranged concentrically about the central axis Ax, and are disposed spaced apart from each other at equal intervals in a radial direction. The fins **26a**, **26b**, and **26c** are configured to have diameters such that the diameter of the fin **26a** is smallest, the diameter of the fin **26b** is larger than the diameter of the fin **26a**, and the diameter of the fin **26c** is largest.

(31) The fins **26a**, **26b**, and **26c** are configured such that the respective lengths by which the fins **26a**, **26b**, and **26c** extend from a base **23a** of the fin member **23** are longer toward an inner fin. That is, the length by which the fin **26** extends toward the pipe **21b** is longest for the innermost fin **26a**, followed by the middle fin **26b**, and the outermost fin **26c** in this order. Although three fins **26** are illustrated in FIGS. 2, 3A, and 3B, the number of fins are not limited to this, and one, two, or four or more fins may be provided.

(32) Note, however, that the fins **26a**, **26b**, and **26c** may be configured such that the respective lengths of extension from the base **23a** of the fin member **23** are the same or such that the respective lengths of extension from the base **23a** of the fin member **23** are longer toward an outer fin. Alternatively, the middle fin **26b** may be configured to be longer or shorter than the innermost fin **26a** and the outermost fin **26c**.

(33) In the fin member **23** according to such a configuration, the length by which each fin **26** extends from the base **23a** of the fin member **23** is designed to be an appropriate length such that the exhaust gas led out from the outlet **32** of the pipe **21b** is able to hit the fins **26a**, **26b**, and **26c**. That is, the fins **26a**, **26b**, and **26c** extend toward the outlet **32** while being spaced apart from the outlet **32** by an appropriate distance. Hence, the exhaust gas led out from the outlet **32** hits the base **23a** of the fins **26a**, **26b**, and **26c** at a substantially right angle, thus creating a flow of the exhaust

gas in the fin member **23**. The flow is created by turning back the exhaust gas in the fin member **23** in the opposite direction to the direction of the flow of the exhaust gas in the first exhaust path **31** of the pipe **21b**. As a result, the exhaust gas led out from the outlet **32** hits the fin member **23** and is turned back inside of the fin member **23** in the direction of the second exhaust path **33**, thus causing the exhaust gas to flow into the second exhaust path **33**.

(34) The exhaust gas is heated by a heater while being passed through the exhaust gas introduction pipe **21** and is led out from the outlet **32**. The exhaust gas that is exhausted is cooled inside of the fin member **23** and by a trap surface **27**, which is the inner surface of the exhaust path forming member **22** forming the second exhaust path **33**. The inside of the fin member **23** and the trap surface **27** are cooled by a refrigerant such as a coolant flowing in the flow path **24a**. As a result, the byproducts contained in the exhaust gas resolidify, and subsequently adhere to and are deposited on the surfaces of the fins **26a**, **26b**, and **26c**, and the trap surface **27** of the exhaust path forming member **22**. This allows the byproducts contained in the exhaust gas to be resolidified and trapped. In FIG. 2, the byproducts trapped by the trap device **20** are indicated as byproducts R. Note, however, that the entrapment locations of the byproducts R illustrated in FIG. 2 are merely examples.

(35) Particularly, in this embodiment, the exhaust gas sufficiently hits the fin member **23** that extends in the opposite direction to the direction of the flow of the exhaust gas led out from the outlet **32**, thus creating a flow of the exhaust gas in the fin member **23**. As a result, the resolidified byproducts R can adhere more easily to the surface of the plurality of fins **26a**, **26b**, and **26c**, and the entrapment efficiency of the byproducts R contained in the exhaust gas can be increased. Furthermore, the exhaust gas is turned back inside of the fin member **23** and flows into the second exhaust path **33**, thus causing the resolidified byproducts R to be deposited on the trap surface **27** when the exhaust gas flows through the second exhaust path **33**. An example of the result of such a process is illustrated in a cross-sectional view taken along a plane B-B of FIG. 2 in FIG. 4. As illustrated in FIG. 4, the byproducts R contained the exhaust gas are trapped by the trap surface **27** of the exhaust path forming member **22**.

(36) The byproducts R contained in the exhaust gas are resolidified, are trapped by the trap surface **27** while passing through the second exhaust path **33**, are turned back again upon hitting a surface **28** (see FIG. 2) of the housing **25** facing the fin member **23**, and flow through the third exhaust path **34**. The inner surface of the housing **25** and the outer surface of the exhaust path forming member **22**, which form the third exhaust path **34**, and the surface **28** may have protrusions **39** formed thereon. The formation of the protrusions **39** increases the surface areas of the housing **25** and the exhaust path forming member **22** that are cooled by the cooling jacket **24**, thereby increasing the cooling efficiency of the exhaust gas and promoting the resolidification of the byproducts R contained in the exhaust gas.

(37) The exhaust gas that flowed through the third exhaust path **34** passes through slits **36a** and flows into a fourth exhaust path **35** formed perpendicular to the third exhaust path **34**. The exhaust gas changes its flow to a direction perpendicular to the third exhaust path **34** along a surface **29** of the housing **25** where the exhaust vent **37** is formed, and the exhaust gas flows through the fourth exhaust path **35** toward the inner portion (the center) of the surface **29**. The surface **29** includes a circular opening at its center, and connects to the exhaust vent **37** through slits **36b** provided at the opening. The exhaust gas that flowed through the fourth exhaust path **35** is exhausted from the exhaust vent **37** via the slits **36b**.

(38) According to the trap device **20** of the first embodiment described above, the exhaust gas that flows through the first exhaust path **31** hits the base **23a** of the fins **26** at a substantially right angle and is turned back inside the fins **26** to flow in the opposite direction of the flow of the first exhaust path **31** to flow into the second exhaust path **33**. This configuration allows the exhaust gas to flow into the deep portions of the fins **26**, thus creating a flow of the exhaust gas inside of the fins **26**. Hence, the amount of byproducts R trapped by the fin member **23** can be increased. As a result, the

efficiency by which the byproducts R are trapped by the trap device **20** can be improved. Furthermore, by providing a configuration in which the exhaust gas that is turned back by being passed through the fin member **23** flows through the second exhaust path **33**, the entrapment efficiency of the trap surface **27** can be improved. As a result, it is possible to increase the life of the trap device **20**.

Second Embodiment

(39) [Trap Device]

(40) The configuration of the trap device **20** according to the second embodiment will be described next with reference to FIGS. **5**, **6A**, and **6B**. FIG. **5** is a vertical cross-sectional view of the trap device **20** according to the second embodiment. FIGS. **6A** and **6B** are views illustrating the fin member **23** included in the trap device **20** according to the second embodiment. FIG. **6A** is a plan view taken along a plane A-A of FIG. **5** of the fin member **23** according to the second embodiment. FIG. **6B** is a perspective view of the fin member **23** according to the second embodiment.

(41) The trap device **20** of the second embodiment differs from the trap device **20** of the first embodiment in only the configuration of the fins **26a**, **26b**, and **26c** of the fin member **23**, and other configurations are the same as those of the first embodiment. Hence, the configuration of the fins **26a**, **26b**, and **26c** will be described below, and a description of other configurations of the trap device **20** will be omitted.

(42) As illustrated in FIGS. **5**, **6A**, and **6B**, the plurality of fins **26a**, **26b**, and **26c** according to the second embodiment are arranged concentrically about the central axis Ax and are disposed spaced apart from each other at equal intervals in a radial direction. The fins **26a**, **26b**, and **26c** according to the second embodiment include cutouts **26a1**, **26b1**, and **26c1**, respectively, each having a comb shape. The cutouts **26a1** form recessed openings in the fin **26a**, the cutouts **26b1** form recessed openings in the fin **26b**, and the cutouts **26c1** form recessed openings in the fin **26c**.

(43) The fins **26a**, **26b**, and **26c** are configured such that the respective lengths by which the fins **26a**, **26b**, and **26c** extend from the base **23a** of the fin member **23** are longer toward an outer fin. That is, the length by which the fin **26** extends from the base **23a** is longest for the outermost fin **26c**, followed by middle fin **26b**, and the innermost fin **26a** in this order.

(44) The cutouts **26a1** include three cutouts that are formed at equal intervals along the circumference of the fin **26a**. The cutouts **26b1** include three cutouts that are formed at equal intervals along the circumference of the fin **26b**. The cutouts **26c1** include three cutouts that are formed at equal intervals along the circumference of the fin **26c**.

(45) As illustrated in FIGS. **6A** and **6B**, the respective cutouts **26a1** and **26b1** of the fins **26a** and **26b** that are next to each other are arranged in an alternate manner so as not to overlap each other. Further, the respective cutouts **26b1** and **26c1** of the fins **26b** and **26c** that are next to each other are arranged in an alternate manner so as not to overlap each other.

(46) Such a configuration allows the exhaust gas led out from the outlet **32** to hit the base **23a** of the fins **26a**, **26b**, and **26c** at a substantially right angle, thereby allowing the exhaust gas to pass through the cutouts **26a1** of the fin **26a**, the cutouts **26b1** of the fin **26b**, and the cutouts **26c1** of the **26c** when the exhaust gas flows into the respective inner portions of the fins **26a**, **26b**, and **26c**. Hence, the exhaust gas can flow into the deep portions of the fin member **23** more easily, and the flow of the exhaust gas inside of the fin member **23** can be further improved.

(47) In addition, to prevent the respective cutouts of the neighboring fins **26a**, **26b**, and **26c** from overlapping each other, the cutouts **26a1**, **26b1**, and **26c1** are arranged in an alternate manner. Such an arrangement allows the exhaust gas that passes through the cutouts **26a1**, **26b1**, and **26c1** to collide with the walls of the neighboring fins **26** more easily. Thus, it is possible to achieve a configuration in which the exhaust gas is caused to actively collide with the cooled fins **26** such that resolidification of the byproducts R contained in the exhaust gas is easily promoted. As a result, the amount of the byproducts R that can be trapped inside of the fin member **23** can be increased, thus improving the entrapment efficiency of the trap device **20**.

(48) Furthermore, the fins **26a**, **26b**, and **26c** are configured such that the respective lengths by which the fins **26a**, **26b**, and **26c** extend from the base **23a** of the fin member **23** are longer toward an outer fin. That is, the length by which the fin **26** extends toward the pipe **21b** is longest for the outermost fin **26c**, followed by the middle fin **26b**, and the innermost fin **26a** in this order. As a result, when the exhaust gas that passed through the cutouts **26a1**, **26b1**, and **26c1** flows inside of the fin member **23**, the exhaust gas can collide more easily with the all of the surfaces of the fins **26a**, **26b**, and **26c**, and thus the entrapment efficiency of the trap device **20** can be improved.

(49) Hence, it is preferable for the fins **26a**, **26b**, and **26c** to be configured such that the respective lengths by which the fins **26a**, **26b**, and **26c** extend from the base **23a** of the fin member **23** are longer toward an outer fin. Note, however, that the respective lengths of the fins **26a**, **26b**, and **26c** are not limited to this. In a similar manner to the first embodiment, the fins **26a**, **26b**, and **26c** may be configured such that the respective lengths by which the fins **26a**, **26b**, and **26c** extend from the base **23a** of the fin member **23** are the same or such that the respective lengths by which the fins **26a**, **26b**, and **26c** extend from the base **23a** of the fin member **23** are longer toward an inner fin. Alternatively, the middle fin **26b** may be configured to be longer or shorter than the innermost fin **26a** and the outermost fin **26c**.

(50) Although three fins are illustrated in FIGS. **5**, **6A**, and **6B** as the plurality of fins **26**, the number of fins are not limited to this, and one, two, or four or more fins may be provided. Further, although it is preferable for all of the fins **26a**, **26b**, and **26c** to include cutouts as in this embodiment, the configuration is not limited to this. It may be configured so that at least one of the fins **26a**, **26b**, or **26c** includes the cutouts. Furthermore, although the fins **26a**, **26b**, and **26c** each include three cutouts in this embodiment, the configuration is not limited to this. The number of cutouts to be included in each of the fins **26a**, **26b**, and **26c** may be the same or may be different among the fins **26a**, **26b**, and **26c**.

(51) According to the trap device **20** of the second embodiment described above, the exhaust gas that flowed through the first exhaust path **31** hits the base **23a** of the fins **26** at a substantially right angle, is turned back within the fins **26** to the opposite direction of the flow in the first exhaust path **31**, and subsequently flows into the second exhaust path **33**. Particularly, in the second embodiment, the flow of the exhaust gas inside of the fin member **23** can be further improved by providing the cutouts **26a1** in the fin **26a**, the cutouts **26b1** in the fin **26b**, and the cutouts **26c1** in the fin **26c**.

(52) Hence, the exhaust gas can flow into the deep portions of the fins **26**, pass through the alternately arranged cutouts **26a1**, **26b1**, and **26c1**, and collide with the walls of the fins **26a**, **26b**, and **26c** such that the amount of the byproducts **R** trapped by the fin member **23** can be further increased. As a result, the efficiency by which the trap device **20** traps the byproducts **R** can be improved. Furthermore, by providing a configuration in which the exhaust gas is turned back by being passed through the fin member **23** from the first exhaust path **31** to flow into the second exhaust path **33**, the entrapment efficiency can be improved. Therefore, the life of the trap device **20** can be further increased.

(53) As described above, according to the trap device **20** of this embodiment, the shape of the fins **26** of the fin member **23** provided at a position facing the exhaust gas introduction pipe **21** can be optimized to improve the efficiency of trapping the byproducts **R** contained in the exhaust gas.

(54) The trap device and the semiconductor manufacturing device according to the embodiments disclosed herein are merely examples in all respects and are not intended to limit the present invention. Various modifications and changes can be made to the embodiments without departing from the scope and the spirit of the appended claims. Aspects described in the above embodiments can be applied to or combined with other configurations to the extent that are not contradictory.

(55) The semiconductor manufacturing device incorporating the trap device according to the present disclosure is applicable to any type of the following devices such as an atomic layer deposition (ALD) device, a capacitively coupled plasma (CCP) device, an inductively coupled

plasma (ICP) device, a radial line slot antenna (RLSA) device, an electron cyclotron resonance plasma (ECR) device, or a helicon wave plasma (HWP) device.

(56) Furthermore, although a plasma processing device has been described as an example of the semiconductor manufacturing device incorporating the trap device according to the present disclosure, the semiconductor manufacturing device is not limited to the plasma processing device as long as it is a device that performs a given process (for example, a deposition process, an etching process, or the like) on a substrate.

(57) The present application is based on and claims priority to Japanese Patent Application No. 2020-107117, filed on Jun. 22, 2020, the entire contents of which are hereby incorporated herein by reference.

DESCRIPTION OF THE REFERENCE NUMERALS

(58) **1** chamber **2** plasma **4** lower electrode **10** semiconductor manufacturing device **20** trap device **21** exhaust gas introduction pipe **22** exhaust path forming member **23** fin member **24** cooling jacket **31** first exhaust path **33** second exhaust path **34** third exhaust path **35** fourth exhaust path **37** exhaust vent **40** controller L exhaust line ST stage R byproduct W substrate

Claims

1. A trap device comprising: an exhaust gas introduction pipe configured to allow an exhaust gas to flow and be led out from an outlet; a fin member provided in a position where the fin member faces the outlet and is hit by the exhaust gas led out from the outlet; an exhaust path forming member covering at least a portion of the exhaust gas introduction pipe and including an exhaust path configured to exhaust the exhaust gas such that the exhaust gas is turned back, via the fin member, in an opposite direction to a direction of flow of the exhaust gas in the exhaust gas introduction pipe; and a cooling jacket configured to cool the fin member, wherein the fin member includes a fin extending in the opposite direction to the direction of the flow of the exhaust gas in the exhaust gas introduction pipe.
2. The trap device as claimed in claim 1, wherein the fin includes a cutout that is comb-shaped.
3. The trap device as claimed in claim 2, further comprising a plurality of fins, the plurality of fins including the fin, wherein the plurality of fins are concentrically arranged, and wherein the plurality of fins are configured such that the respective lengths by which the plurality of fins extend from a base of the fin member are longer toward an outer fin.
4. The trap device as claimed in claim 2, further comprising a plurality of fins, the plurality of fins including the fin, wherein the plurality of fins each include the cutout, and wherein the respective cutouts of the fins that are next to each other are arranged in an alternate manner.
5. The trap device as claimed in claim 2, further comprising a plurality of cutouts, the plurality of cutouts including the cutout, wherein the plurality of cutouts are provided at equal intervals along a circumference of the fin.
6. The trap device as claimed in claim 2, wherein the cutout in the fin forms a recessed portion.
7. The trap device as claimed in claim 1, further comprising a plurality of fins, the plurality of fins including the fin, wherein the plurality of fins are concentrically arranged, and wherein the plurality of fins are disposed spaced apart from each other at equal intervals in a radial direction.
8. The trap device as claimed in claim 1, wherein the fin member is disposed at a position where the exhaust gas flowing in the exhaust gas introduction pipe hits the fin member at a substantially right angle.
9. A semiconductor manufacturing device comprising: a chamber; and a trap device configured to be connected to the chamber, wherein the trap device includes an exhaust gas introduction pipe configured to allow an exhaust gas to flow and be led out from an outlet, a fin member provided in a position where the fin member faces the outlet and is hit by the exhaust gas led out from the outlet, an exhaust path forming member covering at least a portion of the exhaust gas introduction

pipe and including an exhaust path configured to exhaust the exhaust gas such that the exhaust gas is turned back, via the fin member, in an opposite direction to a direction of flow of the exhaust gas in the exhaust gas introduction pipe, and a cooling jacket configured to cool the fin member, wherein the fin member includes a fin extending in the opposite direction to the direction of the flow of the exhaust gas in the exhaust gas introduction pipe.
